

FACULTY OF SCIENCE, ENGINEERING AND ENVIRONMENT



ASSESMENT REPORT (NATURAL LANGUAGE PROCESSING)

Topic Modelling Starting with Manually Collected Dataset of
Unstructured Text

SUBMITTED BY

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1. INTRODUCTION

The rise of cryptocurrencies and blockchain technology has revolutionized the financial landscape, sparking widespread discussion, speculation, and innovation. Online communities, particularly on platforms like Reddit, serve as hubs for enthusiasts, investors, and developers to exchange ideas, analyze market trends, and debate the future of digital assets. Understanding the key themes within these discussions can provide valuable insights into market sentiment, technological advancements, and emerging risks in the cryptocurrency ecosystem.

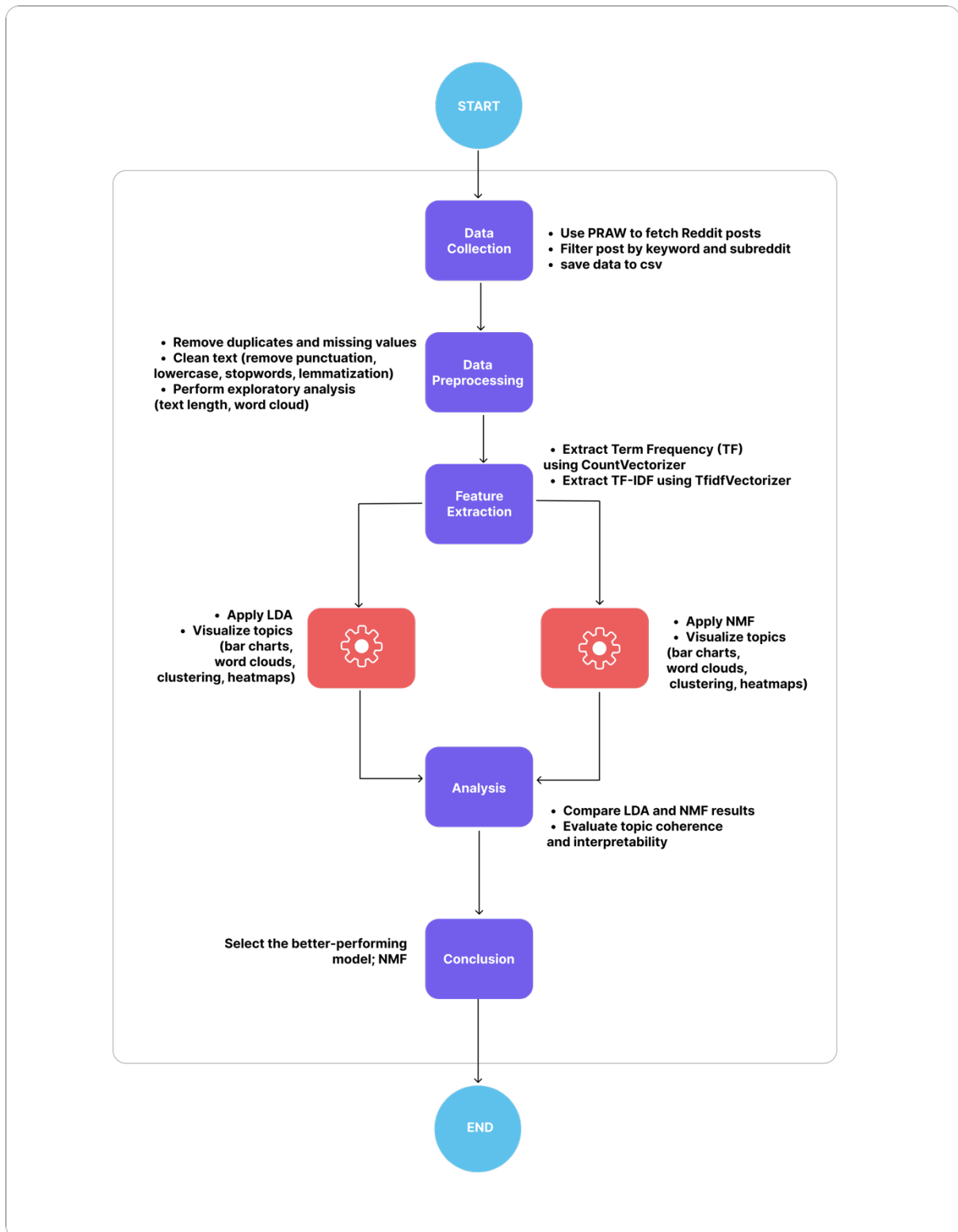
This report presents a topic modeling analysis of cryptocurrency-related discussions from Reddit, leveraging two powerful techniques Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF) to uncover dominant themes and patterns. By examining posts from subreddits such as r/CryptoCurrency, r/Bitcoin, r/Ethereum, and others, we identify the most frequently discussed topics, ranging from price speculation and trading strategies to blockchain technology, regulations, and decentralized finance (DeFi).

The objectives of this study are:

- To extract and categorize the main discussion topics in cryptocurrency forums.
- To compare the effectiveness of LDA and NMF in modeling short-text discussions.
- To visualize topic distributions, word associations, and thematic clusters for better interpretability.

By combining statistical topic models with interactive visualizations, we transform raw forum data into actionable knowledge about the cryptocurrency landscape. The following sections detail our methodology, present comparative results between LDA and NMF implementations, and discuss implications for various stakeholders in the digital asset ecosystem, we aim to shed light on the priorities and concerns of the cryptocurrency community, offering a data-driven perspective on trends that shape this rapidly evolving market. The findings can help stakeholders make informed decisions, whether in trading, project development, or policy engagement.

The figure below shows the basic flow of the process:



2. DATASETS

The dataset was collected using the Reddit API via the Python library praw (Python Reddit API Wrapper). The data consists of 1,500 posts from cryptocurrency-related subreddits, including: **r/CryptoCurrency**, **r/Bitcoin**, **r/Ethereum**, **r/Altcoin**, **r/NFT**, **r/Finance**, **r/Trading**, **r/Investment** with search criterias that include: **cryptocurrency**, **crypto**, **blockchain**, **bitcoin**, **ethereum**, **business**, **finance**, **trading**, **wallet**, **digital**, **economy**, **stocks**, **currency**.

2.1 Searching for Related Posts

To fetch related posts with other informations like title, created time etc. The followed steps were taken:

1. **Get an API Credentials:** I needed to create a developers account to fetch my client id, secret key, useragents so as to enable communication with Reddit API.
2. **Using Search Function:** I implemented the reddit API search function and passed my subreddit and keywords to the API.

```
# Set up the Reddit API client
reddit = praw.Reddit(client_id='jNDebfJyUZYLz6TA9BFiw', client_secret='8kNu2dL5bDQLmfCYXH4DmIa4wUTFzQ', user_agent='developers')

# Define the keywords and subreddits to search in
keywords = ['cryptocurrency', 'crypto', 'blockchain', 'bitcoin', 'ethereum', 'business', 'finance', 'trading', 'wallet', 'digital']
subreddits = ['CryptoCurrency', 'Bitcoin', 'Ethereum', 'Altcoin', 'NFT', 'Memecoins', 'Finance', 'Trading', 'Investment'] # Specific

# Define the number of posts to retrieve in total
num_posts = 1500

# Define a list to store the extracted data
data = []

# Loop through each subreddit
for subreddit in subreddits:
    try:
        # Use PRAW's search method with a limit
        search_results = reddit.subreddit(subreddit).search(f"{' OR '.join(keywords)}", limit=num_posts)
        for result in search_results:
            # Check if the post has a non-empty selftext field
            if result.selftext:
                # Extract the relevant data from each post
                post_title = result.title
                post_text = result.selftext
                post_author = result.author.name if result.author else "Unknown"
                post_created_utc = result.created_utc
                # Store the data in the list
                data.append({'title': post_title, 'text': post_text, 'author': post_author, 'created_utc': post_created_utc})
                print(f"Retrieved post {len(data)} from subreddit {subreddit}: {post_title}")
            # Stop if the desired number of posts is reached
            if len(data) >= num_posts:
                break
        # Delay to avoid hitting rate limits
        time.sleep(4)
    except Exception as e:
        print(f"Error fetching posts from subreddit {subreddit}: {e}")
```

3. EXPLORATORY DATA ANALYSIS

3.1 Data Inspection

Using the function “.head()” we able to view the first few rows then the columns of the datasets.

```
Exploratory Analysis

posts_df.head()
```

	title	text	author	created_utc
0	A list of 43 countrys and their cryptocurrency...	It took a bit more time than expected but i co...	steavus	1.616069e+09
1	Messari's "Crypto Theses for 2024" (with summa...	If you don't know who they are, Messari is a p...	jawni	1.704224e+09
2	Will Bitcoin replace the US Dollar in our life...	The history and growth of Bitcoin in the last ...	nobjos	1.637755e+09
3	The definitive list of non-crypto crypto inves...	With the influx of crypto users, I decided to ...	JazzyJayKarr	1.615045e+09
4	How To Get Rich With Bitcoin Even If You Have ...	**The surprising** **wealth-building secret*...	Sweet-Count2557	1.668180e+09

3.2 Data Description

The datasets is initially made up of 1197 entries in each column with no null value.

```
posts_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1197 entries, 0 to 1196
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   title       1197 non-null   object
1   text        1197 non-null   object
2   author      1197 non-null   object
3   created_utc 1197 non-null   float64
dtypes: float64(1), object(3)
memory usage: 37.5+ KB
```

To ensure the datasets is clean, consistent and ready for further analysis, the following operation were performed:

1. Removing Duplicates:

- Eliminates duplicate rows to ensure each row is unique.

2. Handling Missing Values:

- Drops rows with missing (NaN) values to maintain data integrity.

3. Ensuring Unique Titles and Text:

- Groups the data by the title column and keeps the first occurrence of each title.
- Further groups by the text column to ensure unique post content.

4. Resetting the Index:

- Resets the DataFrame index to a default integer-based index for consistency.

5. Data Summary:

Generates descriptive statistics for all columns to verify the balance and quality of the cleaned dataset.

```
# Drop duplicate rows
posts_df = posts_df.drop_duplicates()

# Drop rows with missing values
posts_df = posts_df.dropna()

# Group by 'title' to ensure unique titles
posts_df = posts_df.groupby('title').first().reset_index()
posts_df = posts_df.groupby('text').first().reset_index()

# Reset the index
posts_df = posts_df.reset_index(drop=True)

# Verify the balance
posts_df.describe(include='all')
```

✓ 0.1s Python

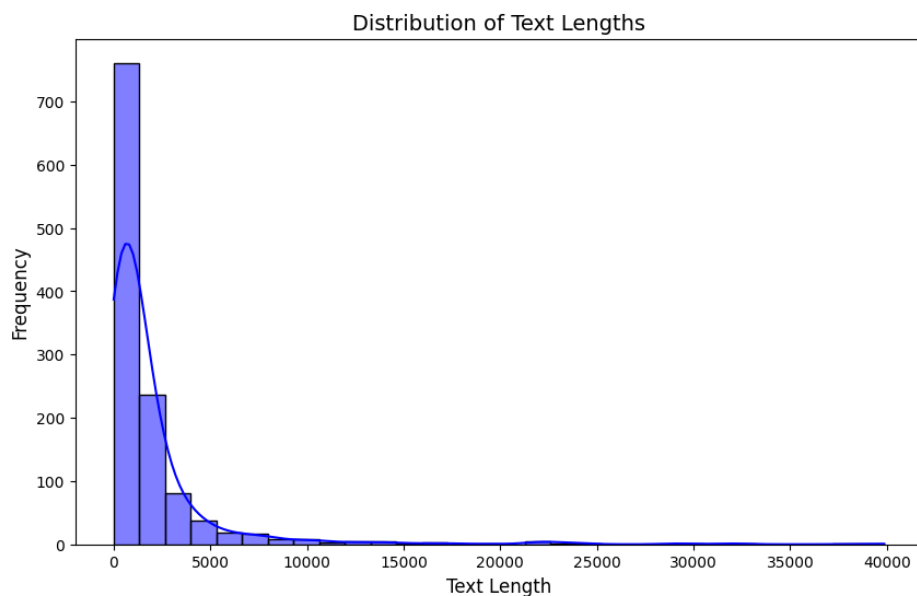
	text	title	author	created_utc
count	1170	1170	1170	1.170000e+03
unique	1170	1170	948	NaN
top	🚀 Mission-Driven: Protecting the Crypto Space\...	Catbat Will Be The Next Billion-Dollar Crypto-...	DAHUDMJ	NaN
freq	1	1	18	NaN
mean	NaN	NaN	NaN	1.693557e+09
std	NaN	NaN	NaN	6.205148e+07
min	NaN	NaN	NaN	1.271341e+09
25%	NaN	NaN	NaN	1.640137e+09
50%	NaN	NaN	NaN	1.725250e+09
75%	NaN	NaN	NaN	1.738240e+09
max	NaN	NaN	NaN	1.745420e+09

Generate Code Markdown

```
posts_df = posts_df.drop(columns=['created_utc', 'author'])
```

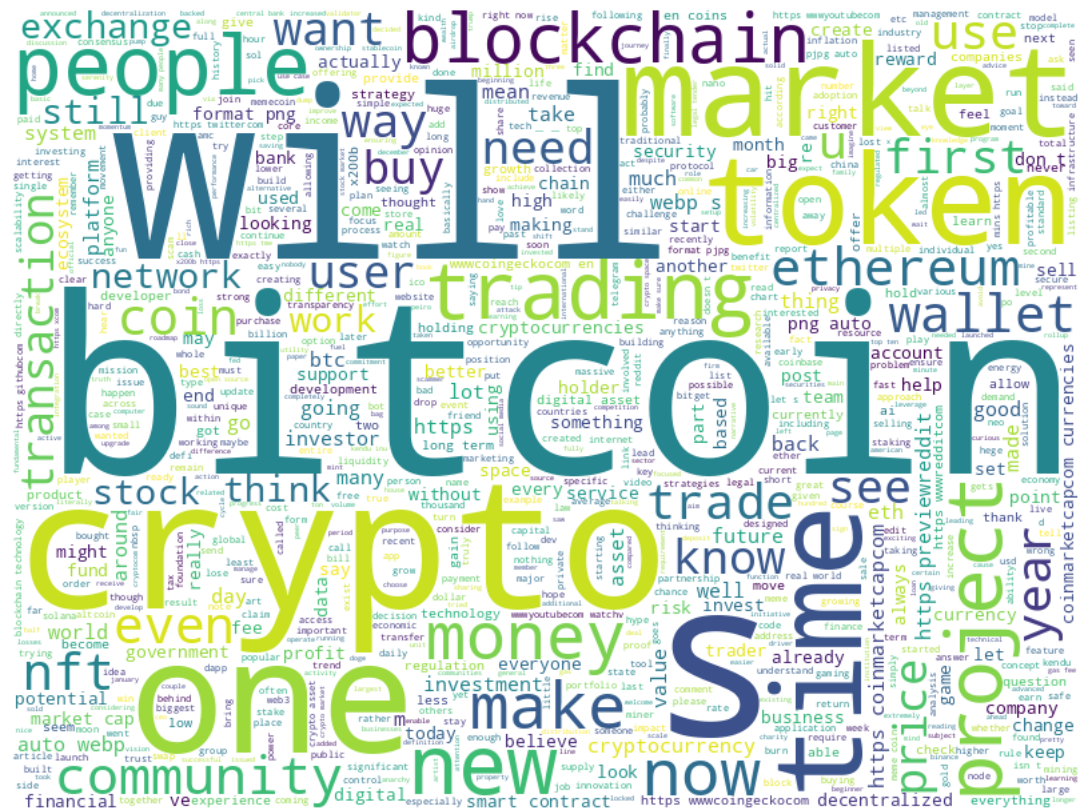
✓ 0.0s Python

3.3 Text Length: I used matplotlib to generate a histogram that illustrates the distribution of text lengths within the texts which shows the text length spans from 0 to 5,000 characters.



Word Cloud

Let's display the most common words in our texts dataset.



The most common words in the word cloud, such as "bitcoin", "crypto", "token", "will", "blockchain", "one" and "time" suggest these are the most frequently mentioned or emphasized terms in corpus. The presence of words such as "market," "trading," "user," "people," "project," "nft," and "money" indicate discussions related to the cryptocurrency, possibly trading, investing in new projects.

Data Preprocessing

1. Stopwords Removal:

- Uses NLTK's English stopwords list to filter out common words.
- The **remove_stopwords()** function processes the text to exclude these words.

2. Text Cleaning:

- **sent_to_words()** converts sentences into lowercase word lists, removing punctuation.
- Applies this to a DataFrame column (**cleaned_text**) containing post content.

3. DataFrame Processing:

- Creates a new column **text_without_stopwords** with stopwords removed from the original text.
- Adds a **text_lemmatized** column using WordNet lemmatization to reduce words to their base forms.

4. Output:

- Prints the first 30 words of the first processed document as a sample.
- Shows the final DataFrame structure with the new processed columns.

The code transforms raw text into a cleaner format suitable for LDA analysis by removing noise (stopwords/punctuation) and normalizing words (lemmatization).


```
# Drop duplicate rows
posts_df = posts_df.drop_duplicates()

# Drop rows with missing values
posts_df = posts_df.dropna()

# Group by 'title' to ensure unique titles
posts_df = posts_df.groupby('title').first().reset_index()
posts_df = posts_df.groupby('text').first().reset_index()

# Reset the index
posts_df = posts_df.reset_index(drop=True)

# Verify the balance
posts_df.describe(include='all')
```

✓ 0.1s Python

	text	title	author	created_utc
count	1170	1170	1170	1.170000e+03
unique	1170	1170	948	NaN
top	🚀 Mission-Driven: Protecting the Crypto Space ... Catbat Will Be The Next Billion-Dollar Crypto-...	DAHUDMJ		NaN
freq	1	1	18	NaN
mean	NaN	NaN	NaN	1.693557e+09
std	NaN	NaN	NaN	6.205148e+07
min	NaN	NaN	NaN	1.271341e+09
25%	NaN	NaN	NaN	1.640137e+09
50%	NaN	NaN	NaN	1.725250e+09
75%	NaN	NaN	NaN	1.738240e+09
max	NaN	NaN	NaN	1.745420e+09

🔗 Generate + Code + Markdown

```
posts_df = posts_df.drop(columns=['created_utc', 'author'])
```

✓ 0.0s Python

```
# Remove stopwords
posts_df['text_without_stopwords'] = posts_df['text'].apply(lambda x: ' '.join([w for w in x.split() if w.lower() not in stopwords])

# perform lemmatization
posts_df['text_lemmatized'] = posts_df['text_without_stopwords'].apply(lambda x: ' '.join([wn.lemmatize(w) for w in x.split() if w

# take a look at the data
posts_df.head()
```

✓ 5.8s Python

	text	title	cleaned_text	text_without_stopwords	text_lemmatized
0	\n\n\nOverview: The ICON Project (ICX) is a cu...	The ICON Project (ICX): Revolutionizing Blockc...	\n\n\noverview: the icon project (icx) is a cu...	Overview: ICON Project (ICX) cutting-edge bloc...	Overview: ICON Project (ICX) cutting-edge bloc...
1	\n\n\nSender AI Network is taking blockchain t...	Sender Ai aims to improve the blockchain	\n\n\nsender ai network is taking blockchain t...	Sender AI Network taking blockchain next level...	Sender AI Network taking blockchain next level...
2	\n\n1. Which is the stable currency to hold as...	🌟which currency to hold in Assets ?🚀	\n\n1 which is the stable currency to hold as ...	1. stable currency hold asset long-term , cons...	1. stable currency hold asset long-term , cons...
3	\n\nEdit 8: Important **NEW TOOL**--> [Coinba...	This can't be put off or delayed. They are abo...	\n\nedit 8: important **new tool**--> [coinba...	Edit 8: Important **NEW TOOL**--> [Coinbase c...	Edit 8: Important **NEW TOOL**--> [Coinbase c...
4	\n\nFirst things first, trading for a living i...	80% Yearly Returns Trading Big Caps - My Basic...	\n\nfirst things first trading for a living is...	First things first, trading living going take ...	First thing first, trading living going take t...

4. MODEL IMPLEMENTATION

In this section, The way the data was prepared using CountVectorizers, TF-IDF (Term Frequency – Inverse Document Frequency) and how Latent Dirichlet Allocation (LDA), Non-Negative Matrix Factorization (NMF) models were used for our topic modelling.

Feature Extraction Count & TF-IDF Vectorizer

```
Prepare Text for Training

# assign values to parameters
n_topics = 5
n_top_words = 10

✓ 0.1s Python

print("Extracting tf features for LDA...")
tf_vectorizer = CountVectorizer(max_df=0.90, min_df=4, max_features=1000)
tf = tf_vectorizer.fit_transform(posts_df['text_lemmatized'])
print("Extracted tf features for LDA.")
✓ 0.5s Python

Extracting tf features for LDA...
Extracted tf features for LDA.

print("Extracting tf-idf features for NMF...")
tfidf_vectorizer = TfidfVectorizer(max_df=0.90, min_df=4, max_features=1000)
tfidf = tfidf_vectorizer.fit_transform(posts_df['text_lemmatized'])
print("Extracted tf-idf features for NMF.")
✓ 0.2s Python

Extracting tf-idf features for NMF...
Extracted tf-idf features for NMF.
```

- **max_df (maximum document frequency):** Set to 0.90, this excludes words that appear in more than 90% of documents. This helps filter out overly common words (e.g., generic terms) that don't contribute meaningfully to topic differentiation.
- **min_df (minimum document frequency):** Set to 4, this removes words that appear in fewer than four documents. This eliminates rare terms that could introduce noise or lead to overfitting in the topic model.
- **max_features:** Limits the vocabulary to the top 1,000 most frequent words in the corpus. This focuses the analysis on the most significant terms while reducing computational overhead.

These settings help refine the dataset for LDA by balancing relevance and efficiency—removing both overly common and overly rare words while keeping the most informative terms.

Model Implementation of LDA and NMF

```
# build LDA model with tf
lda = LatentDirichletAllocation(n_components=n_topics, random_state=42)
lda.fit_transform(tf)
tf_feature_names = tf_vectorizer.get_feature_names_out()
```

```
plot_top_words(lda, tf_feature_names, n_top_words, "Topics in LDA model\n")
```

✓ 14.9s

Python

NMF Model Training

```
nmf = NMF(n_components=n_topics, random_state=42, beta_loss="frobenius").fit(tfidf)
tfidf_feature_names = tfidf_vectorizer.get_feature_names_out()
```

```
plot_top_words(nmf, tfidf_feature_names, n_top_words, "Topics in NMF model")
```

✓ 3.5s

Python

LDA (LatentDirichletAllocation)

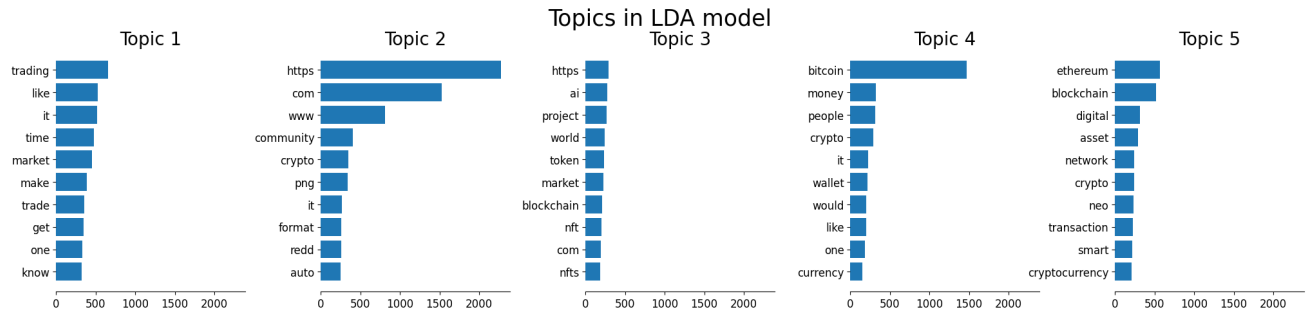
- Model Initialization:
 - LatentDirichletAllocation is a probabilistic model that discovers latent topics in text.
 - n_components=n_topics: Defines the number of topics to extract.
 - random_state=42: Ensures reproducibility.
- Training & Transformation:
 - fit_transform(tf): Fits the model to term-frequency (TF) vectors (raw word counts) and returns document-topic distributions.
- Feature Names:
 - tf_feature_names: Retrieves the vocabulary (words) from the TF vectorizer.
- Visualization:
 - plot_top_words(): Displays the top n_top_words per topic, helping interpret themes.

NMF(Non-Negative Matrix Factorization)

- Model Initialization:
 - NMF is a linear algebra-based model that factorizes the term-document matrix.
 - beta_loss="frobenius": Uses Frobenius norm for optimization (suitable for TF-IDF).
- Training:
 - fit(tfidf): Fits the model to TF-IDF vectors (weighted word counts, emphasizing rare but important terms).
- Feature Names:
 - tfidf_feature_names: Gets the vocabulary from the TF-IDF vectorizer.
- Visualization:
 - plot_top_words(): Shows the most relevant words per topic.

RESULTS AND EVALUATION

Topics extracted using LDA:



Topics

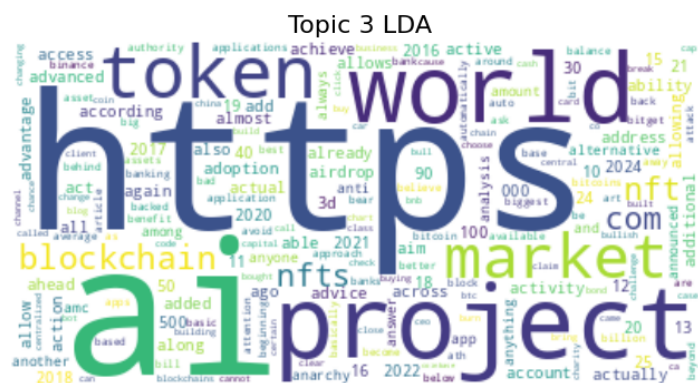
“trading”, “like”, “it”, “time”,
“market”, “make”, “trade”,
“get”, “one”, “know”



“https”, “com”, “www”,
“community”, “crypto”,
“png”, “it”, “format”, “redd”,
“auto”



“https”, “ai”, “project”,
“world”, “token”, “market”,
“blockchain”, “nft”, “com”,
“nfts”



**“ethereum”, “blockchain”,
“digital”, “asset”, “network”,
“crypto”, “neo”,
“transaction”, “smart”,
“cryptocurrency”**

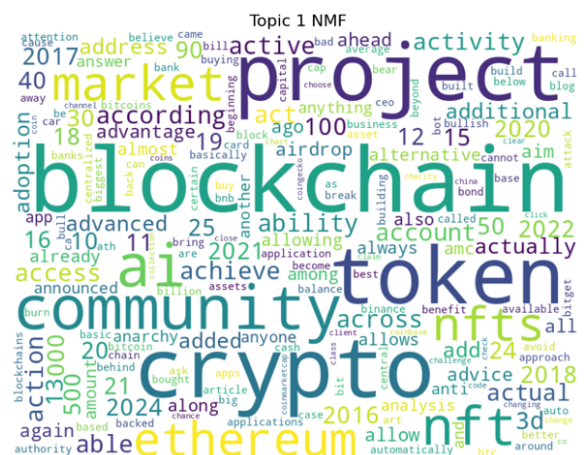
A word cloud visualization of terms related to cryptocurrency and digital finance. The most prominent words are "bitcoin", "crypto", "wallet", "people", "like", "money", "would", "active", "advanced", "achieve", "it", "one", "cross", "according", "adoption", "ability", "actual", "amount", "app", "alternative", "almost", "also", "able", "access", "again", "account", "all", "allow", "aim", "airdrop", "attack", "balance", "back", "action", "aggravating", "age", "ago", "ahead", "algorithm", "analysis", "analogous", "another", "anyone", "application", "applied", "applies", "always", "built", "better", "2020", "along", "2021", "future", "forecast", "2024", "3d", "act", "13g", "19", "add", "allow", "aim", "around", "16", "additional", "current", "allows", "already", "advance", "activity", "attention", "buy", "city", "annual", "18", "bit", "24", "secure", "set", "sell", "series", "share", "sharpened", "short", "single", "small", "some", "soon", "start", "step", "stock", "store", "strategy", "successful", "superior", "switch", "take", "talk", "target", "tax", "technology", "technical", "terms", "than", "that", "the", "there", "they", "this", "those", "through", "thus", "to", "too", "towards", "under", "until", "upon", "us", "we", "with", "within", "without", "witness", "yet", "you", "your", "yours", "yourself", "yourselves".

[illegible]

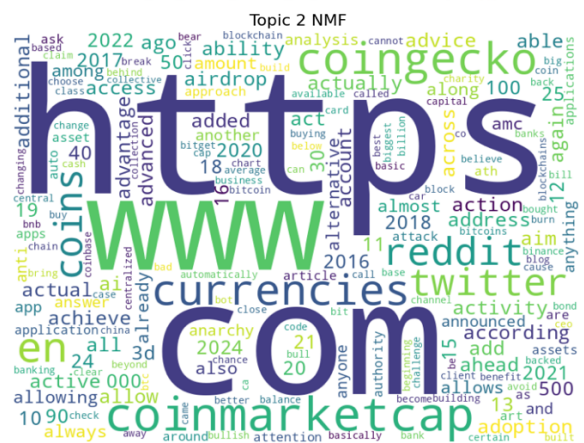
Figure 10 displays five horizontal bar charts, each representing the top 10 topics for a specific NMF model (Topic 1 through Topic 5). The x-axis for each chart represents the topic score, ranging from 0.0 to 1.5. The y-axis lists the topics. The topics are ranked by their association with the model.

- Topic 1:** blockchain, crypto, project, token, community, ai, nft, market, ethereum, nfts.
- Topic 2:** https, com, www, coinmarketcap, coingecko, currencies, en, reddit, twitter, coins.
- Topic 3:** bitcoin, btc, currency, people, crypto, money, wallet, buy, price, it.
- Topic 4:** png, redd, webp, width, auto, format, preview, it, https, pipg.
- Topic 5:** trading, trade, know, time, money, would, like, day, trader, get.

**“blockchain”, “crypto”, “project”,
“token”, “community”, “ai”, “nft”,
“market”, “ethereum”, “nfts”**



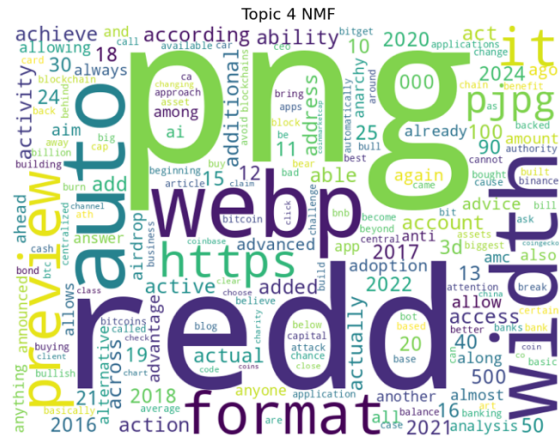
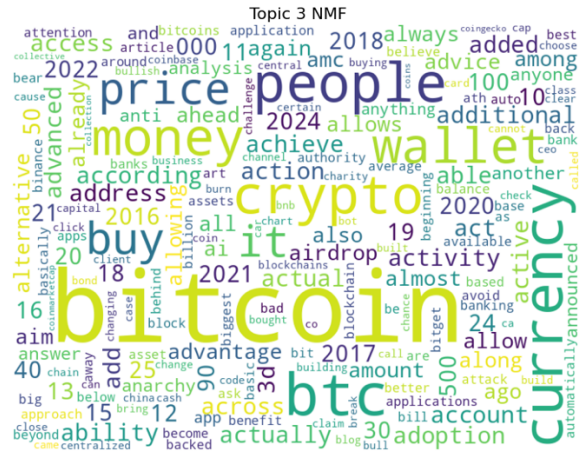
**“https”, “com”, “www”,
“coinmarketcap”, “coingecko”,
“currencies”, “en”, “reddit”, “twitter”,
“coins”**



**“bitcoin”, “btc”, “currency”, “people”,
“crypto”, “money”, “wallet”, “buy”,
“price”**

**“png”, “redd”, “webp”, “width”, “auto”,
“format”, “preview”, “it”, “https”,
“pjpg”**

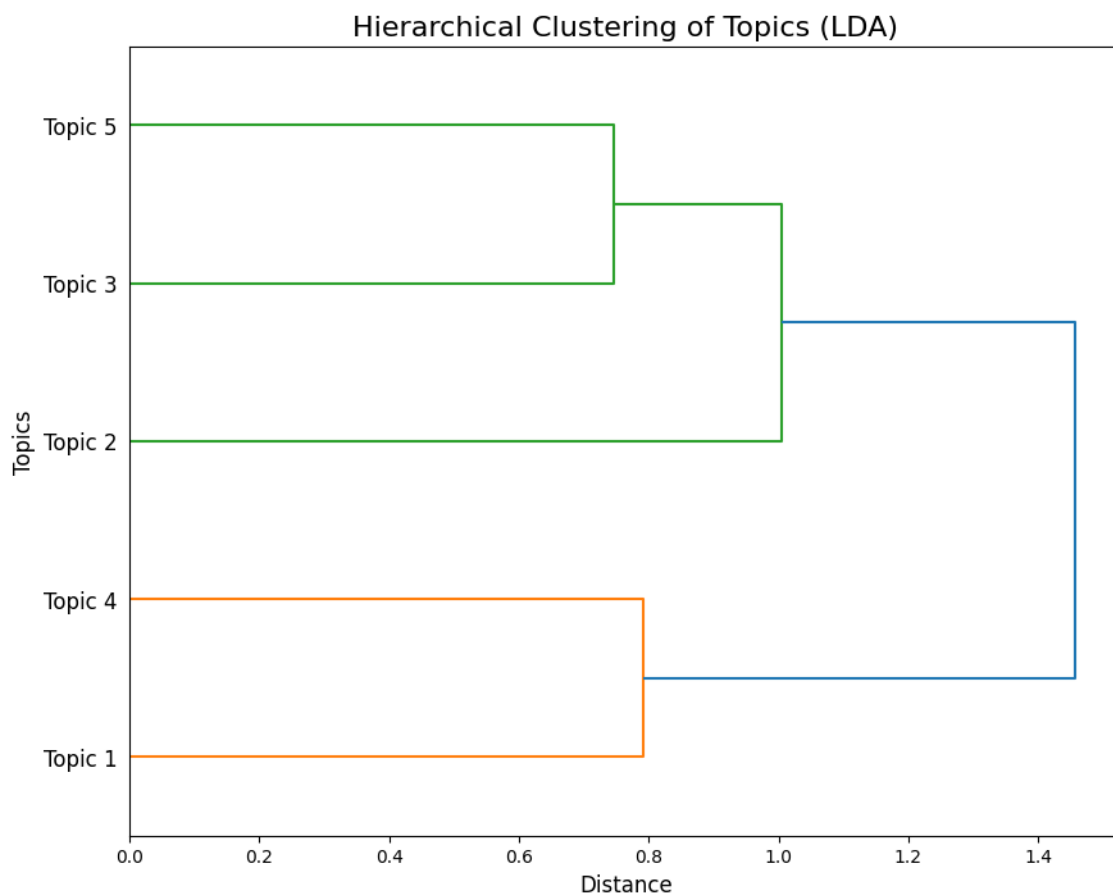
**“trading”, “trade”, “know”, “time”,
“money”, “would”, “like”, “day”,
“trader”, “get”**



Hierarchical Clustering

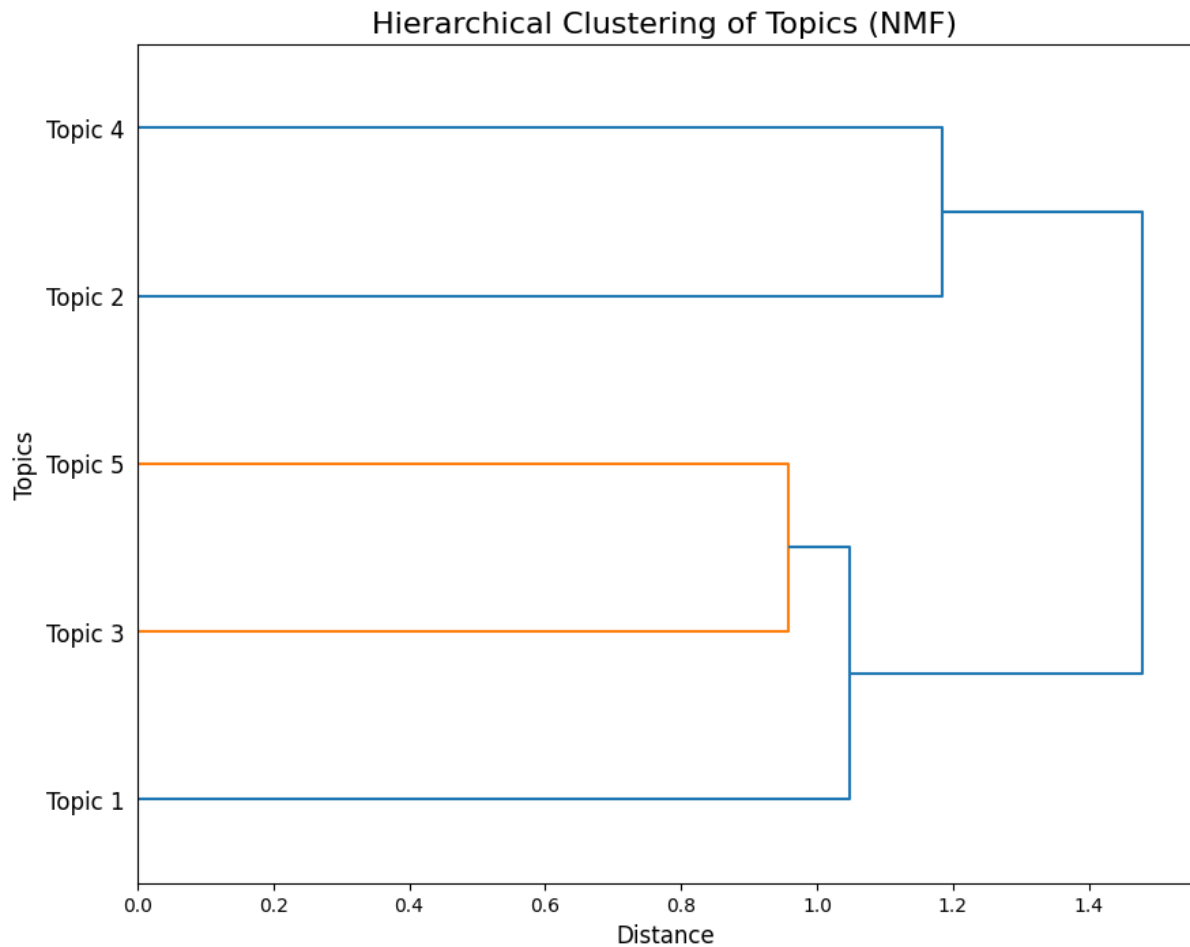
In Hierarchical Clustering, the height at which branches merge indicates how similar or distinct the clusters (or topics) are. Topics that are closely related (similar in content) will merge at lower heights, appearing closer together. Topics that are very different will merge much higher up, appearing farther apart. This visual representation helps quickly identify which topics share common themes and which stand apart.

LDA Model



Topic 1 and Topic 4 are paired together indicating a strong similarity just like Topic 3 and Topic 5 which are also paired together. Topic 2 stands on its own which joins Topic 3 and 5 at a higher Level. Whereas Topic 1 and Topic 4 join the other topics at the highest level which means all topics connected in one way or the other.

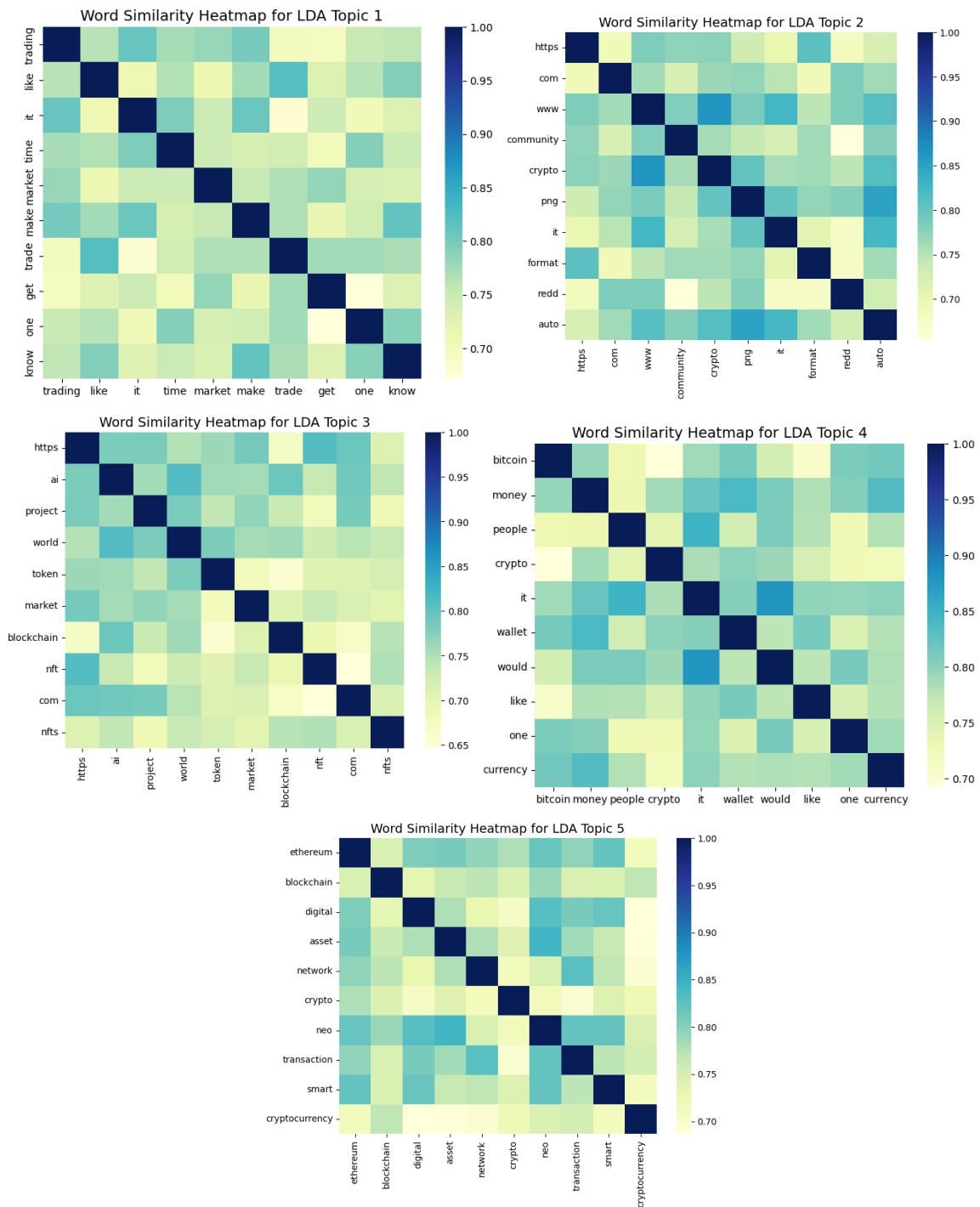
NMF Model



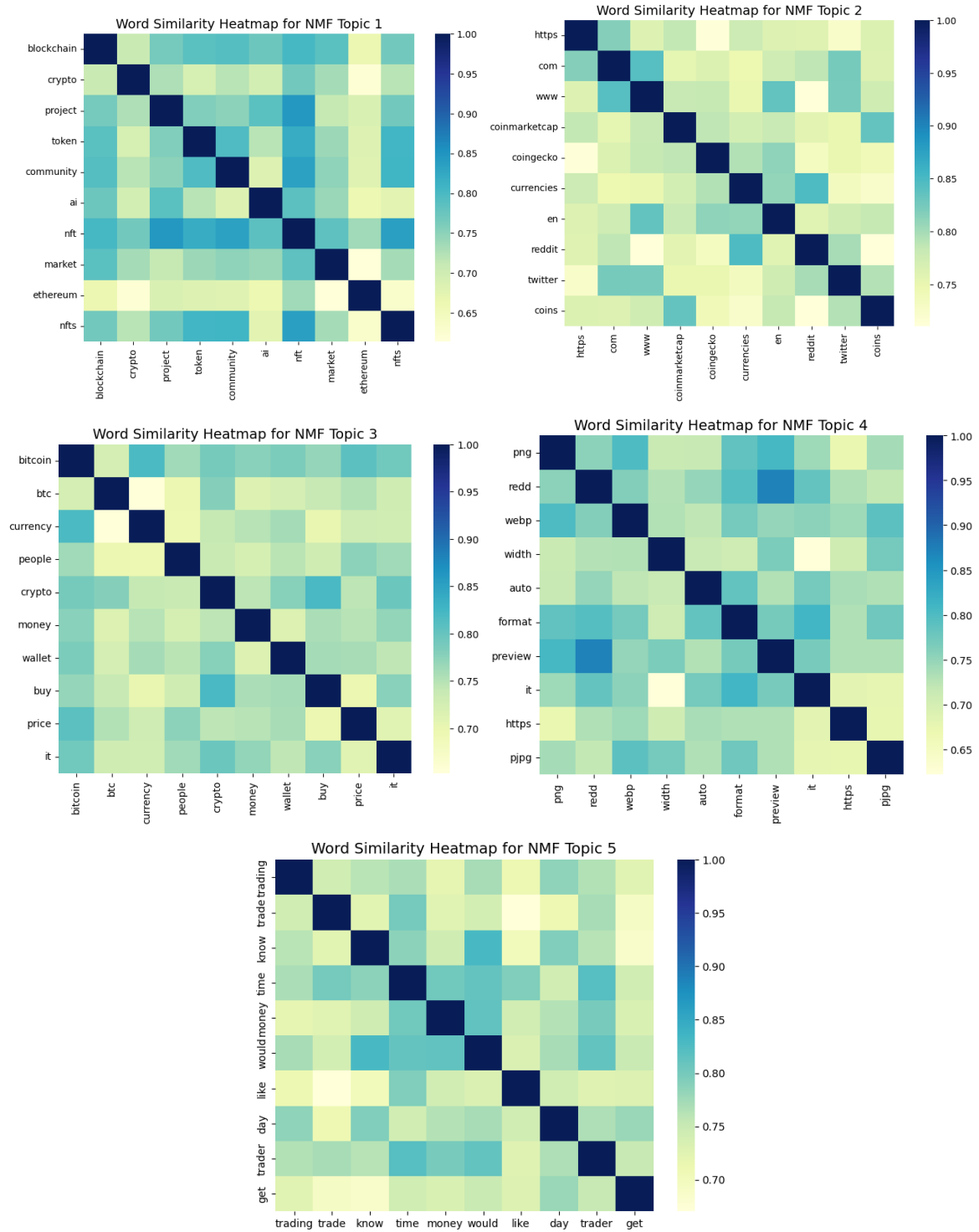
Topic 3 and 5 are paired together this time in NMF just like Topic 2 and 4 but with a higher level which means they have close similarities. Topic 1 stands alone joining Topic 3 and 5 at a high level of 0.1 then proceeds to join Topic 2 and 4 at the highest level still showing the connections between topics. The overall height of connection at which all the topics merge implies a diversity in the topics related to Cryptocurrency. Each topic represents a different approach and indicates different perspectives.

Heat Maps

LDA Model



NMF Model



CONCLUSION

Based on the results of the topic modeling implementation, Non-Negative Matrix Factorization (NMF) appears to have performed better than Latent Dirichlet Allocation (LDA) in this specific context. Here's a detailed comparison and reasoning.

The analysis compared the performance of NMF and LDA for topic modeling across several key aspects. NMF demonstrated superior topic coherence, with its topics showing closely related words that clearly represented specific themes, likely due to its use of TF-IDF which emphasizes meaningful but less frequent terms. In contrast, while LDA produced interpretable topics, it exhibited more overlap between them, suggesting less distinct boundaries, a limitation attributed to its probabilistic nature.

Visualizations like word clouds reinforced these findings, with NMF displaying sharper thematic focus in its word distributions compared to LDA's slightly more diffuse representations. Hierarchical clustering and heatmaps further highlighted NMF's advantages, revealing better-defined clusters and stronger word relationships within topics, whereas LDA's clusters appeared less distinct with some topics blending together.

NMF proved more interpretable, with its topics aligning clearly with expected themes, making it preferable for applications requiring actionable insights, while LDA remained viable but required more effort to disentangle overlapping themes.

While both models successfully extracted meaningful topics, NMF outperformed LDA in terms of topic coherence, interpretability, and distinctiveness. This suggests that NMF is better suited for this dataset, particularly because the TF-IDF representation highlights important but less frequent terms, which aligns well with the nature of the Reddit posts.

For future work, consider:

- 1. Hyperparameter Tuning:** Experimenting with the number of topics (`n_topics`) and vectorizer parameters (`max_df`, `min_df`) to further optimize results.
- 2. Hybrid Approaches:** Combining NMF and LDA insights to leverage the strengths of both models.
- 3. Domain-Specific Embeddings:** Using pre-trained embeddings like Word2Vec or GloVe to enhance topic coherence further.

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Alameer, A. (n.d.). NLP Repository. GitHub. Retrieved from <https://github.com/Ali-Alameer/NLP>